

Smaran Vallabhaneni

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EDUCATION

Olin College of Engineering <i>Bachelor of Science, Engineering in Computing (GPA: 4.0)</i>	Needham, MA Aug. 2025 – May 2029
– Relevant Courses: Linear Algebra, Multi-Variable Calculus, Differential Equations, Modeling & Simulation, Bayesian Statistics, Artificial Intelligence, Machine Learning, Software Design (Python), Kernel Development (C), Algorithms and Data Structures, Cryptography, Data Science, Statistics for Data Science, Data Analysis with Python, Data Preparation & Applied Machine Learning, Predictive Business Analytics, Products & Markets, Principles of Integrated Engineering, Introduction to Sensors, Instrumentation & Measurement	

SKILLS

Programming Languages: C, C++, Rust, Java, Python, R, MATLAB, NumPy, pandas, MySQL, NoSQL, TypeScript, JavaScript, C#, Swift, Dart, HTML, CSS, Bash, $\LaTeX$
Frameworks: PyTorch, Scikit-Learn, LLMs, RAG Systems, Ollama, OpenCV, MediaPipe, ROS, MAVROS, React, Flutter, SwiftUI, Django, AngularJS, Kivy
Tools: GitHub Copilot, Google Gemini, Google Antigravity, Claude Code, LM Studio, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Node.js, Next.js, Tableau Desktop, Docker, ArduPilot, Figma, Adobe Creative Suite
Operating Systems: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based)
Other: HuggingFace, NLP, Amazon Web Services, Google Cloud, Microsoft Azure, DevOps, API Integration, Testing and Debugging, Continuous Integration Pipelines, UI, UX
Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail

PROJECTS

W.A.G.E <i>Python, TypeScript, Next.js, FastAPI, MySQL, Ollama</i>	Jul. 2026
– Architected a live job-board feed NLP inference pipeline; postings are tokenized, chunked, and scored via nomic-embed-text embeddings and chatgpt-oss-20b semantic similarity, with sub-5-second end-to-end latency across 25+ concurrent documents.	
– Backed by a MySQL-stored FastAPI REST layer for privacy-preserving text extraction; optional Ollama backend swaps cloud inference for local execution with identical query semantics.	
Specter <i>Python, Linux, Ollama</i>	May 2026
– Designed a real-time vision-language inference pipeline on Linux/Wayland; routes live screen state to a local gemma4 vision model, achieving 3-second action latency and 86% task classification accuracy across diverse UI environments.	
3D Gesture Tracking <i>Python, OpenCV, MediaPipe, Blender</i>	Apr. 2026
– Implemented a real-time hand landmark detection and gesture classification pipeline via MediaPipe and OpenCV; achieves sub-50ms end-to-end latency and 95% precision (5% FPR) across varied lighting conditions and hand geometries.	
Whoa-Scope <i>Python, Kivy</i>	Feb. 2026 – Apr. 2026
– Profiled and restructured heap allocation patterns in a real-time signal visualization tool, reducing peak memory consumption by 200+ MB; stabilized operation on memory-constrained lab hardware and drove organic adoption across 40 concurrent users.	
ClassicTunes <i>Swift, SwiftUI, Metal API, SQLite</i>	Oct. 2025 – Jul. 2026
– Engineered a native macOS GPU rendering pipeline via Metal API shaders and command encoders (120 FPS CoverFlow on 150+ albums); implemented iPod syncing via reverse-engineered binary protocol with custom bit manipulation schema and AES-128 encryption over a SQLite device-state schema.	

EXPERIENCE

CLOS (Command Line Operating System) <i>Founding Engineer and Principal Architect</i>	Remote Apr. 2023 – Present
– Founded CLOS, an offline-first mathematical computing platform with ~1,500 weekly active users; supports graphing, symbolic calculus, simulation, logic design, and AI tutoring across Windows, macOS, and Linux.	
– Engineered 8 core subsystems in Java from scratch: custom DB engine, symbolic math parser, scripting language, 3D renderer, terminal, matrix library, logic gate simulator, and symbolic calculus processor; achieves 60 FPS rendering and sub-50ms latency on multi-variable calculus.	
Olin College AERO Project Team <i>Software Engineering Lead</i>	Needham, MA Jul. 2026 – Present
– Leading a 5-engineer software sub-team across the full autonomy stack (image recognition, stitching, flight planning, autopilot, and ground station via ArduPilot, ROS, MAVROS, NVIDIA Jetson) for AUVSI sUAS; achieved sub-100ms round-trip telemetry latency and stable 20 FPS at 10-mile field range.	
– Drive bi-weekly cross-functional integration cycles with mechanical and electrical sub-teams across 15–20 engineers to synchronize airframe iterations with software requirements.	
Geeky Insights LLC <i>Mobile Platform Engineer, Intern</i>	Remote Apr. 2024 – Apr. 2025
– Designed end-to-end UI/UX for a customizable audio alarm clock app in Figma and Adobe Creative Suite; implemented full front-end and backend in Flutter, Dart, and Java, producing a 15 MB optimized release build.	
– Optimized application memory layout for Android 9+ low-resource devices, reducing RAM consumption to match the baseline footprint of Android’s native system clock.	

ACTIVITIES

Olin ICPC <i>Competitive Programmer</i>	Needham, MA Feb. 2026 – Present
– Solving high-difficulty algorithmic problems under strict time constraints using segment trees, Fenwick trees, DP, graph algorithms (network flow, SSSP, MST), string algorithms, and computational geometry.	

CERTIFICATIONS & PUBLICATIONS

Meta Database Engineer Certification <i>Credential ID: M8WKYG9PMT4G</i>	Jul. 2025
C++ Memory Safety and Integrity Evaluation <i>DOI: 10.58445/rars.2876</i>	Aug. 2025
Performance Evaluation of C++ and Java for Portable Devices <i>DOI: 10.58445/rars.1783</i>	Oct. 2024